HARSHAL RAAJ PATNAIK

Senior Full-Stack & Cloud Systems Engineer | Technical Lead & Platform Architect

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EDUCATION

Indian Institute of Management Kozhikode (IIMK) **2025 – Present**

Executive Post Graduate Programme in Management (EPGPM) – Business Strategy, Platform Economics & Operations *Kochi, India*

SRM Institute of Science and Technology **2017 – 2021**

Bachelor of Technology in Computer Science and Engineering (CGPA: 8.76 / 87.6%) *Chennai, India*

PROFESSIONAL EXPERIENCE

Tata Communications Ltd. **Jul 2021 – Present**

Software Development Engineer (Cloud Infrastructure, Distributed Workflows & Full-Stack) *Chennai, India*

- Slashed cloud batch scheduler execution latency by **92.6% (from ~18 minutes down to 80 seconds)** across distributed Cassandra clusters by implementing strict 50-item batch chunking, multi-level Redis caching, and tuning HikariCP connection pools.
- Architected multi-hypervisor orchestration engine and Flowable workflows across **KVM, VMware ESXi, and OLVM**, managing automated VM provisioning, lifecycle operations, and state compensation via asynchronous **RabbitMQ** queues.
- Built full-stack onboarding workflow for **Caringo Swarm S3-compatible cloud object storage** and automated cPanel DNS A-record associations, enabling **100% self-service cloud storage and public IP lifecycle management**.
- Automated perimeter network security controls via **FortiGate REST APIs** for firewall rules, VDOM tokens, and IPsec VPNs; engineered an in-process sliding-window rate limiter that **eliminated 100% of HTTP 429 rate-limiting failures**.
- Designed dynamic route discovery, wildcard path forwarding, and multi-tenant workspace access validation across **Apache APISIX, Kong, and Tyk** API gateways, integrating downstream audit logging and metering pipelines.
- Engineered switch-port utilization and network capacity telemetry engine integrating Zabbix and Nimsoft APIs, computing cluster-wide CPU, memory, NSX, and datastore allocation to prevent capacity exhaustion.
- Spearheaded frontend modernization to **Angular 17** with reactive component architectures, and led backend modernization from **Java 8 to 17 / Spring Boot 3**, resolving security CVEs in Netty, Log4j, and Apache Commons.
- Stepped up as **Acting Scrum Master** and technical mentor for junior engineers, leading architecture reviews, code reviews, and dependency tracking; earned **3× Extra Mile Commitment Awards** for operational excellence.

STRATEGY & TECHNICAL PRODUCT CAPSTONE

SthiraX: Distributed CO_2-Mineralization & IoT MRV Platform | *IIM Kozhikode Capstone (BPP)* **In Progress**

- Designed an asset-light platform retrofitting regional precast concrete plants with modular industrial CO_2-curing units to address India's **April 2026 C&D Waste Management Rules (mandating 5% to 25% recycled material by 2030 across 20,000+ sqm projects)**.
- Architected batch-level **IoT MRV (Measurement, Reporting & Verification) software** tracking calcium carbonate uptake per production run and generating cryptographically verifiable Digital Product Passports for developer EPR compliance.
- Modeled multi-tier unit economics (equipment leasing, per-tonne curing royalties, and MRV SaaS subscriptions) viable independent of voluntary carbon credit market volatility, supported by a geospatial regional scoring algorithm.

TECHNICAL & LEADERSHIP COMPETENCIES

Languages & Core: Java (8/11/17 LTS), Python, TypeScript, JavaScript, SQL, HTML5/CSS3, Bash
Distributed Systems & Cloud: Spring Boot 3, Flowable BPMN, RabbitMQ (AMQP), Microservices Architecture, REST APIs, KVM, VMware ESXi, OLVM, Docker
Databases, Caching & Storage: Cassandra, Redis, MySQL, Caringo Swarm / S3, Hibernate/JPA, HikariCP (Connection Pool Tuning)
Networking, Security & Gateways: Apache APISIX, Kong, Tyk, FortiGate REST APIs, IPsec VPN, DNS Automation, Sliding-Window Rate Limiting
DevOps, SRE & Observability: Jenkins CI/CD, JaCoCo, JUnit, Mockito, Zabbix APIs, Linux, Maven, SonarQube, Postman, Swagger/OpenAPI
Product Strategy & Leadership: Technical Mentorship, Code Reviews, Agile / Scrum Master, Platform Business Models, Unit Economics, EPR Mandates

HONORS, CERTIFICATIONS & EXTRACURRICULAR

- **3× Extra Mile Commitment Award** – Tata Communications Ltd. for critical cloud platform delivery and reliability.
- **Flipkart Grid 2.0 Engineering Challenge** – Team Lead; achieved **Top 150 Nationwide Rank** out of 20,000+ teams.
- **Architecting with Google Compute Engine** (Google Cloud / Coursera) & **Full Stack Web Development** (Coding Ninjas).
- **Rank 2404 Globally** in Second Round of 5th SOF International Mathematics Olympiad (Science Olympiad Foundation).